

Muhammad Bilal

Senior Frontend Engineer · Angular · React · React Native · AI Interfaces

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🐙 github.com/bilalmughal2000 📍 Islamabad, Pakistan · Remote-ready

3+ yrs Shipping production web & mobile UIs	6+ Production platforms as frontend owner
9–10 AI models streamed in parallel in one UI	5k+ lines Diff viewer rendered without dropped frames

◆ PROFESSIONAL SUMMARY

Senior frontend engineer with 3+ years designing and shipping fast, responsive web applications and cross-platform mobile experiences in **Angular, React and React Native**. I own frontend architecture end-to-end — turning dense product requirements into clean, accessible, high-performance interfaces. Much of my recent work is AI-native: a multi-model workspace that streams 9–10 LLMs in parallel, and a platform that generates and deploys full enterprise applications from plain-English prompts. I pair strong engineering judgment (state architecture, performance budgets, accessibility) with a growing DevOps skill set — Terraform, workload scaling and deploy management — and a genuine obsession with the details users actually feel: speed, clarity and polish.

◆ PROFESSIONAL EXPERIENCE

Trillo Inc. Jun 2023 – Present Islamabad · Onsite / Remote

AI-powered application platform & Google Cloud Premier Partner — turns plain-English requirements into deployable enterprise apps.

Senior Frontend Engineer

Own frontend architecture across several AI-native products, from streaming generation UIs to an IDE-style developer workbench, in Angular 18 with NgRx and Signals.

- ◆ Designed a **streaming-first UI** where multi-step LLM generation (schema, APIs, workflows, agents) renders progressively, with optimistic step previews and a full-screen diff viewer handling **5,000+ line outputs without dropping a frame**.
- ◆ Built a **multi-model AI workspace** that broadcasts a single prompt to **9–10 LLMs in parallel** (Claude, GPT, Gemini and more) as live side-by-side chats, with dictation, transcription and multi-language input.
- ◆ Engineered an **IDE-style backend workbench** — split panes, live logs and a virtualized console holding tens of thousands of entries — later reused to launch a File Manager product.
- ◆ Delivered interactive Angular components and coordinated complex state with **NgRx for run lifecycle and Signals for ephemeral state**.
- ◆ Shipped **cross-platform mobile apps in React Native**, refining UX across iOS and Android.
- ◆ Reduced load and runtime latency through **lazy loading, route-level code splitting and OnPush** change detection.
- ◆ Handle **DevOps** tasks — scaling workloads up/down, updating Terraform files and Python scripts, and managing deploy versions.
- ◆ Own **accessibility (WCAG)** and cross-browser compatibility; partner with design and backend on API contracts and interface refinement.

Web & mobile development across the company product and departmental sites.

Software Engineer

Delivered and maintained web and mobile products, owning key features end-to-end and collaborating across teams.

- ◆ Managed and optimized the company website (**PHP / Laravel**) for performance and security.
- ◆ Developed and customized **WordPress** sites for multiple departments, tailoring functionality to each need.
- ◆ Maintained and upgraded the company's **React Native** mobile app, improving features and performance.
- ◆ Led implementation and optimization of key features for a seamless cross-platform experience.
- ◆ Collaborated with cross-functional teams to define and ship impactful web & mobile features, and proposed new development projects.

◆ SELECTED PROJECTS

Multi-Model AI Workspace

ANGULAR · FRONTEND OWNER

A workspace where a single query fans out to 9–10 AI models at once, rendering each reply as a live streaming chat panel side by side, with hands-free dictation, transcription and multi-language support wired into the backend AI services.

Impact: ask once and compare every model's answer in one view — replacing tab-hopping between separate tools and widening access with voice and multi-language input.

Angular · TypeScript · RxJS · Streaming UI · Web Speech

Trillo AI — ideas to apps in minutes

ANGULAR · FRONTEND ARCHITECTURE

Frontend for a platform that turns plain-English prompts into enterprise apps (auth, schema, APIs, workflows, agents), deployable in one click or exported as full-stack code. Led a streaming-first shell with optimistic previews and a 5k-line diff viewer; NgRx for run lifecycle, Signals for ephemeral state.

Impact: a non-technical founder can ship a working prototype before lunch — validating an idea with zero backend hires.

Angular · TypeScript · NgRx · Signals

AI-Native Business Acquisition Platform

ANGULAR · AGENTIC AI

A unified, role-aware workspace for buyers, brokers and sellers that folds the entire deal lifecycle — discovery, matching, diligence, financing, compliance, close — into one automated pipeline. Built the AI advisor surface (research, drafting, valuation), criteria-based precision matching, and a containerized document-vault UI enforcing zero crossover between deals.

Impact: collapses six fragmented tools into one source of truth for humans and agents; ~100× buyer cost savings and dramatically faster time to close.

Angular · TypeScript · RxJS · Design System

Construction Billing SaaS — AIA G702 / G703

REACT · MAINTENANCE & REDESIGN

Pay-application software with automatic, error-free calculations, a materials-stored summary, smart search and organized project files. Worked across the React frontend — bug fixes, ongoing maintenance and a full theme redesign — sharpening pay-apps, continuation sheets and change orders.

Impact: replaces risky spreadsheets for 2,000+ contractors nationwide; \$50M+ billed monthly with near-zero resubmissions.

React · JavaScript · Theming · Responsive UI

Trillo Backend Workbench

ANGULAR · DEVELOPER TOOLS

IDE-style workbench where developers compose, test and deploy AI-powered workflows. Architected around lazy-loaded feature modules, Signals + OnPush for surgical re-renders and a virtualized log/console layer that stays smooth at tens of thousands of entries.

Impact: desktop-IDE responsiveness at scale; reusable layout primitives let new editor surfaces ship in days and powered a File Manager spin-off.

Angular · Signals · NgRx · Virtualization

AI Document Search Engine

AI SEARCH · CLIMATE TECH

Semantic document search and retrieval for a climate-tech nonprofit — ingestion and chunking, vector embeddings, a semantically indexed retrieval layer and a clean API returning ranked passages with grounded citations, tuned for policy and incentive content.

Impact: compresses hours of manual PDF hunting into seconds of citation-backed semantic search.

Semantic Search · Vector Embeddings · RAG · Retrieval API

Grande Service Departments

REACT NATIVE · FINAL-YEAR PROJECT

A user-friendly room-booking mobile app with one-click booking, real-time availability, intuitive search and a responsive design for seamless access across devices.

Impact: streamlined the entire reservation flow into a single tap; demonstrated advanced React Native skills end-to-end.

React Native · JavaScript · REST APIs

◆ TECHNICAL SKILLS

Frontend Frameworks

Angular 18 · React · React Native · TypeScript · JavaScript ES6+

State & Reactivity

NgRx · Signals · RxJS · Redux

UI, Styling & Data-Viz

HTML5 · CSS3 · Tailwind CSS · Bootstrap · Chart.js · Apache ECharts · Custom SVG

Performance & Accessibility

Lazy loading · Code splitting · OnPush · Virtualization · WCAG / a11y

Integration & AI Interfaces

REST APIs

Axios / Fetch

Firebase

Streaming UI

Web Speech

Multi-model UX

DevOps & Cloud

Terraform

Deploy versioning

Workload scaling

Python scripts

Tooling & Process

Git · GitHub

Bitbucket

Jira / Agile

VS Code

Code review

Foundations

Data Structures

Algorithms

Clean Architecture

Best Practices

◆ EDUCATION

BS Computer Science

International Islamic University Islamabad

2018 – 2022

Intermediate of Computer Science (ICS)

Shiblee Groups of Colleges

2015 – 2017

◆ LANGUAGES

English

Fluent

Urdu

Native

Punjabi

Fluent

◆ CERTIFICATIONS

Deloitte Technology Virtual Experience

Forage · Australia

Aug 2024

Lyft Back-End Engineering Virtual Experience

Forage · United States

Aug 2024

◆ STRENGTHS

Problem-solving

Critical thinking

Communication

Leadership

Adaptability

Fast learner